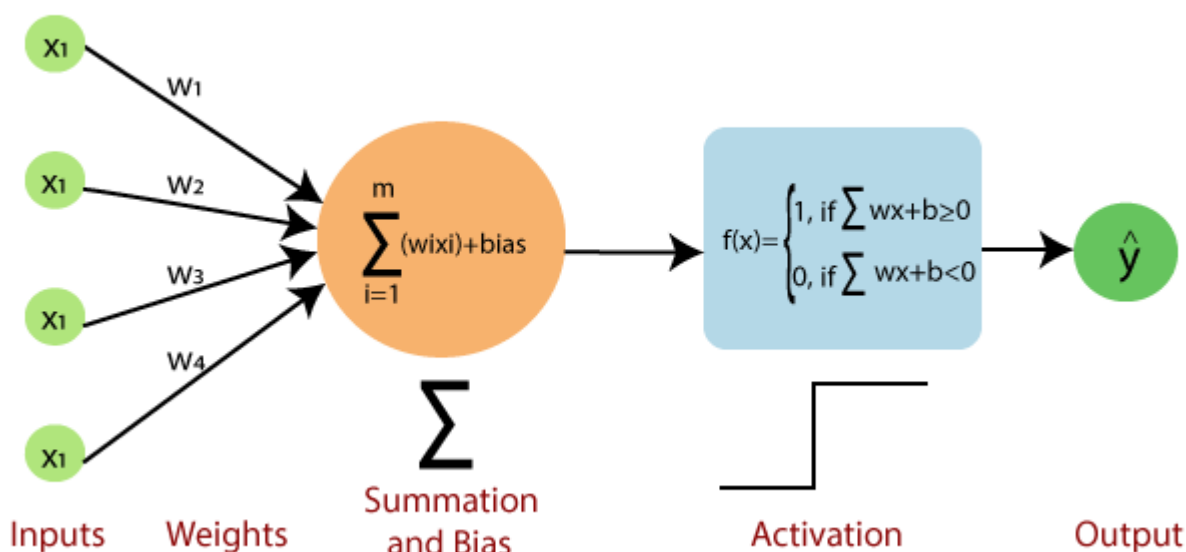


Theory: Perceptron Learning Model

The **Perceptron** is the **simplest form of an Artificial Neural Network** and the foundation of modern machine learning models. It was introduced by **Frank Rosenblatt (1958)** as a computational model inspired by how biological neurons process information.

1. Structure of a Perceptron

A perceptron consists of:



- **Input Layer:** Accepts multiple inputs ($x_1, x_2, \dots, x_n$).
- **Weights (w):** Each input is assigned a weight (w_i) that represents its importance.
- **Summation Function:** Computes the weighted sum
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$$net = \sum_{i=1}^n w_i x_i + b$$

where (b) is the **bias**.

- **Activation Function:** Applies a step or sign function to decide the output:
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$$y = \begin{cases} 1, & \text{if } net \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

2. Learning Rule (Perceptron Training Algorithm)

The perceptron **learns** by adjusting its weights to minimize classification errors. The weight update rule is:

$$w_i^{(new)} = w_i^{(old)} + \eta(t - y)x_i$$

$$b^{(new)} = b^{(old)} + \eta(t - y)$$

Where:

- (η) = Learning rate ($0 < \eta \leq 1$)
- (t) = Target (expected output)
- (y) = Predicted output

Weights are updated repeatedly for all training samples until the error becomes zero or a maximum number of iterations is reached.

3. Working Steps

1. Initialize weights and bias randomly (small values).
 2. For each input pattern:
 - Compute weighted sum.
 - Apply activation function to get output.
 - Update weights using the perceptron learning rule.
 3. Repeat until convergence (no misclassifications).
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4. Example

Used to classify **linearly separable** data like:

- AND, OR logic gates (works correctly).
- Cannot solve XOR (not linearly separable).

5. Advantages

- Simple and easy to implement.
- Converges for linearly separable patterns.
- Forms the basis of multi-layer neural networks.

6. Disadvantages

- Fails for non-linear data (e.g., XOR).
- Fixed learning rate may slow convergence.
- Only binary output (0 or 1).

7. Applications

- Pattern recognition
- Binary classification
- Image and speech recognition (foundation for deep networks)

Conclusion

The **Perceptron Learning Model** demonstrates how a simple neuron can learn to classify input patterns using weight adjustment. Although limited to linearly separable problems, it forms the **building block for modern neural networks** like MLPs and CNNs.